

ECON2043 Experimental and Behavioural Economics

Seminar 6: Problem Set (Projection Bias)

Anru Qiao

Anru.QIAO@nottingham.edu.cn

Juncheng Zhang

Juncheng.ZHANG@nottingham.edu.cn

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Topic Summary

Projection bias

- ▶ People predict future utility using too much of their current state

$$\tilde{u}(c_\tau, s_\tau \mid s_t) = (1 - \alpha)u(c_\tau, s_\tau) + \alpha u(c_\tau, s_t)$$

Different from Time Preferences

- ▶ Time preferences: the mistake is about future self-control or impatience
- ▶ Projection bias: the mistake is about future tastes, moods, or emotions

Question 1

We work through a simple version of the stochastic durable-goods problem in Loewenstein, O'Donoghue, and Rabin (2003).

Suppose you consider purchasing an electronic item on Taobao today ($t = 0$) at price p . It arrives in a few days and only generates utility in one future period, $t = 1$.

Let the future state s_1 describe your valuation of the good. Assume

$$s_1 \in \{0, 1\}, \quad \Pr(s_1 = 0) = \Pr(s_1 = 1) = \frac{1}{2}, \quad \delta = 1$$

- (a) Calculate the expected utility of a rational decision maker ($\alpha = 0$). When does she buy?
- (b) Solve the same problem for a decision maker with projection bias $\alpha = \frac{1}{3}$.
- (c) Comment on when the projection-biased decision maker buys incorrectly.

Answer 1(a): Rational Case

The future expected value of the good is

$$E[s_1] = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 1 = \frac{1}{2}$$

Because $\delta = 1$, expected utility from buying is

$$U_0 = -p + E[s_1] = \frac{1}{2} - p$$

So a rational decision maker buys if and only if

$$\frac{1}{2} - p > 0 \quad \Longleftrightarrow \quad p < \frac{1}{2}$$

Conclusion

- ▶ buy if $p < \frac{1}{2}$
- ▶ do not buy if $p > \frac{1}{2}$
- ▶ indifferent if $p = \frac{1}{2}$

Answer 1(b): Projection Bias

With projection bias,

$$\tilde{U}_0 = E[(1 - \alpha)s_1 + \alpha s_0] - p$$

For $\alpha = \frac{1}{3}$ and $E[s_1] = \frac{1}{2}$,

$$\tilde{U}_0 = \left(1 - \frac{1}{3}\right) \frac{1}{2} + \frac{1}{3}s_0 - p = \frac{1}{3} + \frac{1}{3}s_0 - p$$

So the buying condition is

$$\frac{1}{3} + \frac{1}{3}s_0 - p > 0$$

This depends on the current state:

- ▶ if $s_0 = 1$, then $\tilde{U}_0 = \frac{2}{3} - p$, so she buys if $p < \frac{2}{3}$
- ▶ if $s_0 = 0$, then $\tilde{U}_0 = \frac{1}{3} - p$, so she buys if $p < \frac{1}{3}$

Answer 1(c): When Does She Make a Mistake?

The rational benchmark is: buy if $p < \frac{1}{2}$.

If $s_0 = 1$

- ▶ the biased decision maker buys if $p < \frac{2}{3}$
- ▶ this is too optimistic relative to the rational threshold $\frac{1}{2}$
- ▶ mistake: buys when $\frac{1}{2} < p < \frac{2}{3}$

If $s_0 = 0$

- ▶ the biased decision maker buys only if $p < \frac{1}{3}$
- ▶ this is too pessimistic relative to the rational threshold $\frac{1}{2}$
- ▶ mistake: does not buy when $\frac{1}{3} < p < \frac{1}{2}$

Intuition

- ▶ high current valuation leads to overvaluation of the good: impulse buying and regret
- ▶ low current valuation leads to undervaluation of the good: misses a worthwhile purchase



Thank you very much for joining us!

